BUOYANCY

CONCEPT OF BUOYANT FORCE

Buoyancy is one of those ideas that feels simple (things float or sink) but has a lot of depth once you unpack it properly.

What is Buoyant Force?

Buoyant force is the **upward force exerted by a fluid (liquid or gas)** on an object placed in it. This force exists because fluid pressure increases with depth.

Mathematical Expression of Buoyant Force

$$F_b = \rho_{\text{fluid}} \cdot g \cdot V_{\text{displaced}}$$

Where:

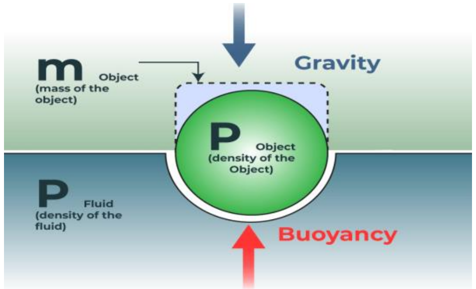
- F_b = buoyant force
- ρ_{fluid} = density of the fluid
- g = acceleration due to gravity
- $V_{\text{displaced}}$ = volume of fluid displaced

☐ ☐ Core Idea: Archimedes' Principle

This principle gives the exact rule:
The buoyant force on an object equals the **weight of the fluid displaced by the object**.
So buoyancy is not random—it depends entirely on how much fluid the object pushes out of the way.

☐ Why Does Buoyant Force Occur?

Fluids exert **pressure in all directions**, and pressure increases with depth:
Top surface of object → lower pressure
Bottom surface → higher pressure
This creates a **net upward force**.



Real-life applications of buoyant force:

☐ **Ships & boats**: Float by displacing water so buoyant force balances weight.
☐ **Submarines**: Control depth using ballast tanks (adjust buoyancy).
☐ **Balloons (hot air/helium)**: Rise due to buoyancy in air.
☐ **Swimming**: Human body floats because water exerts upward force.
☐ **Life jackets**: Increase volume → help a person float.
☐ ☐ **Hydrometers**: Measure liquid density using floating levels.

FLOAT AND SINK

Float and Sink

When an object is placed in a fluid (liquid or gas), it either **floats**, **sinks**, or sometimes stays suspended. This behavior is governed by the interaction between **weight** and **buoyant force**.

Fundamental Principle: Archimedes' Principle

A body immersed in a fluid experiences an upward force equal to the **weight of the fluid it displaces**.

This upward force is called **buoyant force**, and it determines whether the object floats or sinks.

Forces Acting on an Object in a Fluid

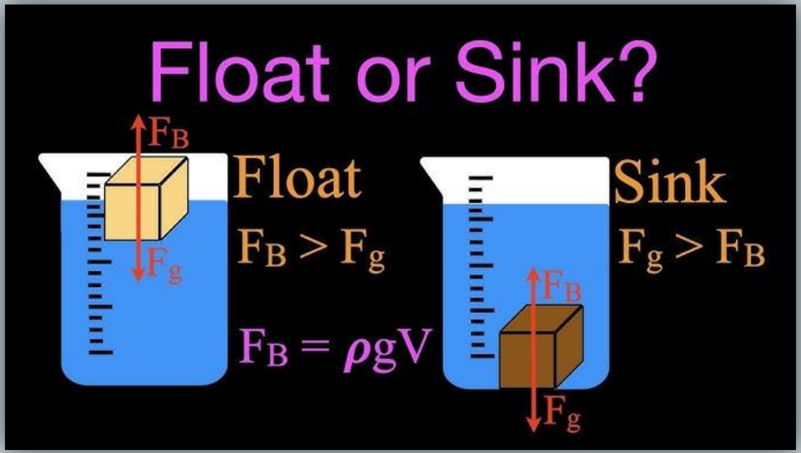
There are two main forces:

- **Weight (W)** → Acts downward

W=m·g

- **Buoyant Force (F_b)** → Acts upward

Fb=ρfluid·g·VdisplacedF



☐ ***FLOATING***

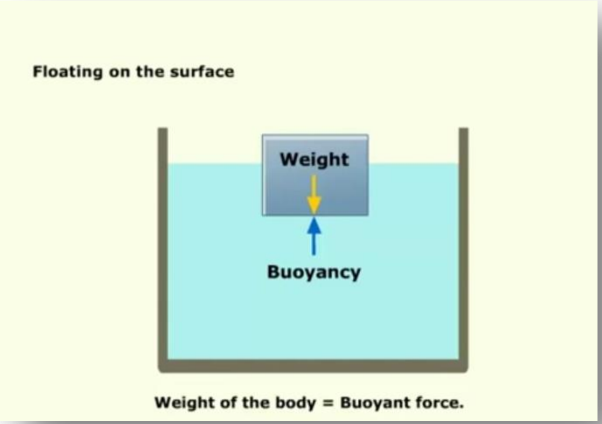
An object floats when: $F_b \geq W$

What actually happens:

Initially, the object may sink a little.

As it sinks, it displaces more fluid → buoyant force increases.

It stops sinking when: $F_b = W$



Key features:

Object is **partially submerged**

Volume adjusts automatically

Seen in low-density objects

Example: A ship made of iron floats because its **overall density (including air inside)** is less than water.

☐ ***SINKING***

An object sinks when: $F_b < W$

What happens:

Even after full submersion, the buoyant force is not enough.

The object continues to move downward until it reaches the bottom.

Key features:

Object is **completely submerged**

High-density objects

Example: A solid iron block sinks in water.

☐ ☐ Role of Density

Density decides float or sink more directly:

If : $\rho_{object} < \rho_{fluid} \Rightarrow$ floats

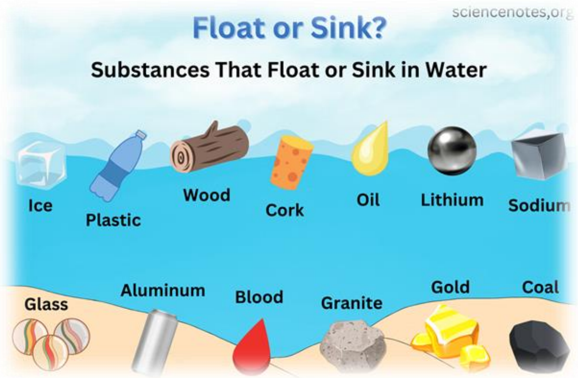
If : $\rho_{object} > \rho_{fluid} \Rightarrow$ sinks

☐ Why Shape Matters

Shape affects **volume and displacement**:
A solid iron block sinks
The same iron shaped into a **hollow ship** floats

☐ Because:

Larger volume → more fluid displaced → greater buoyant force



☐ ☐ Floating in Gases

The same principle applies in air:
Helium balloons float (low density)
Hot air balloons rise when air inside is heated

☐ Real-Life Understanding

You feel lighter in water → buoyant force reduces apparent weight
Ice floats → its density is less than water
Oil floats on water → lower density

☐ Final Summary

Floating and sinking depend on **balance of forces**
Buoyant force depends on **fluid density and displaced volume**
Density comparison gives a quick answer
Shape and structure can change the outcome

ARCHIMEDES PRINCIPLE

When a body is wholly or partially immersed in a fluid, it experiences an **upward buoyant force equal to the weight of the fluid displaced by it**.

When an object is placed in a fluid, it pushes some fluid aside (displacement).

The fluid exerts an **upward force** on the object.

This force depends on how much fluid is displaced—not directly on the object's weight.

☐ **Formula**

$$F_b = \rho \cdot g \cdot V$$

Where:

- F_b = buoyant force
- ρ= density of fluid
- g = acceleration due to gravity
- V = volume of displaced fluid

☐ ☐ **Key Implications**

- **Floating:**

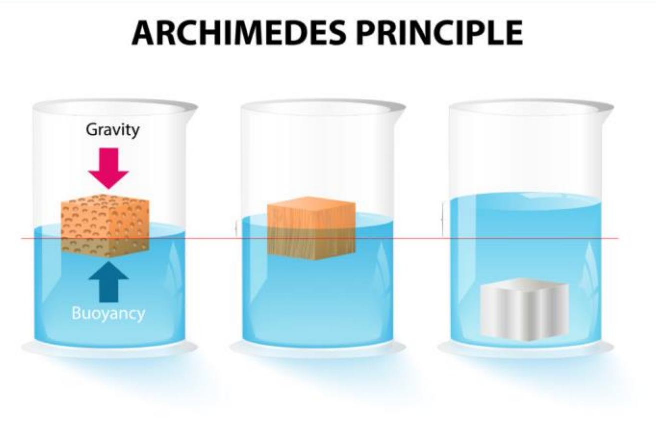
Buoyant force = weight of object

- **Sinking:**

Buoyant force < weight

- **Apparent weight loss in fluid:**

Loss of weight=weight of displaced fluid



Derivation Idea (Conceptual)

Consider an object submerged in a liquid:

Pressure at depth: $P=\rho gh$

Bottom surface → higher pressure

Top surface → lower pressure

Net upward force:

$F_b=(\text{pressure difference})\times\text{area}$

This simplifies to:

$F_b=\rho_{fluid}\cdot g\cdot V_{displaced}$

☐ ☐ **Important Assumptions**

Archimedes' Principle works when:

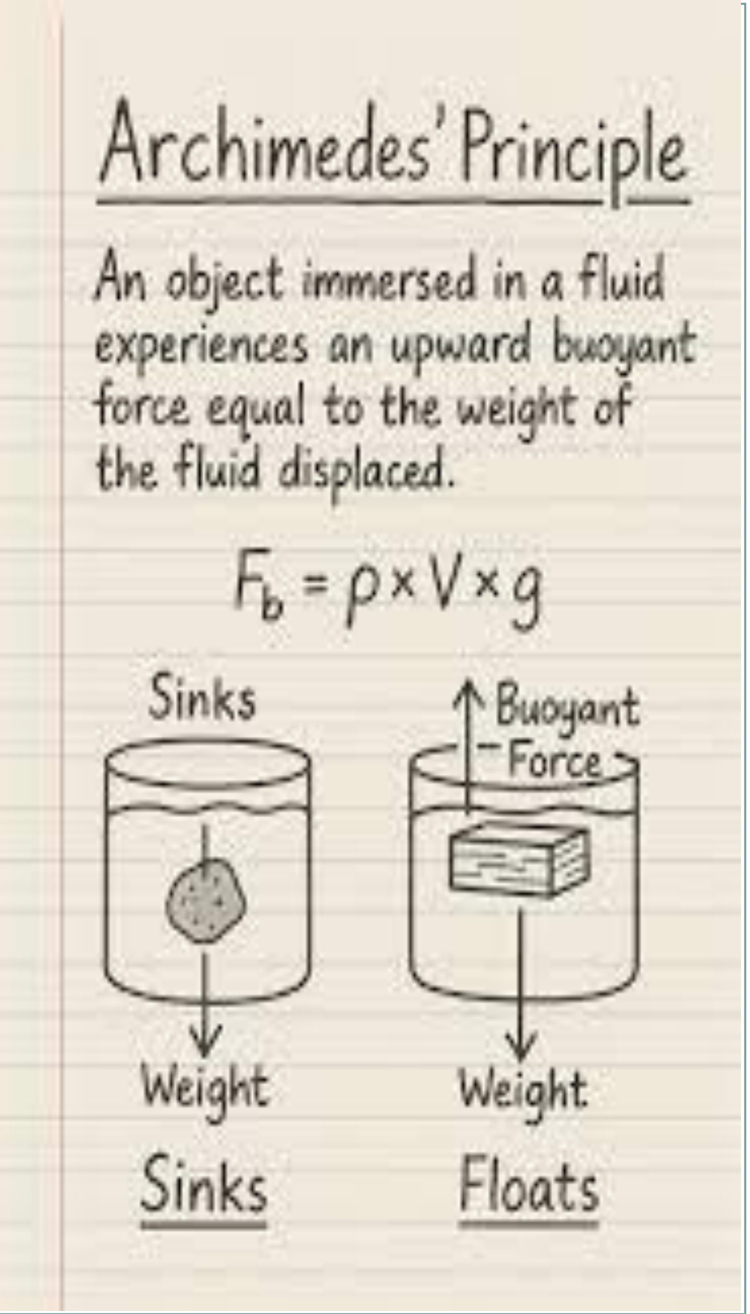
1. Fluid is at rest (no turbulence)
2. Density of fluid is uniform
3. Gravity is constant

☐ ☐ **Applies to Gases Too**

1. Air is also a fluid
2. Balloons rise because they displace heavier air

☐ **Real-Life Importance**

1. Designing ships and submarines
2. Measuring density (hydrometers)
3. Floating bridges and platforms
4. Scuba diving control



□ □ Historical Insight

The principle was discovered by Archimedes when solving a problem about measuring the purity of a crown—leading to the famous "Eureka!" moment.

□ Final Insight

Archimedes' Principle is powerful because it:

Converts a **complex fluid interaction**

Into a **simple measurable quantity** (displaced fluid weight)

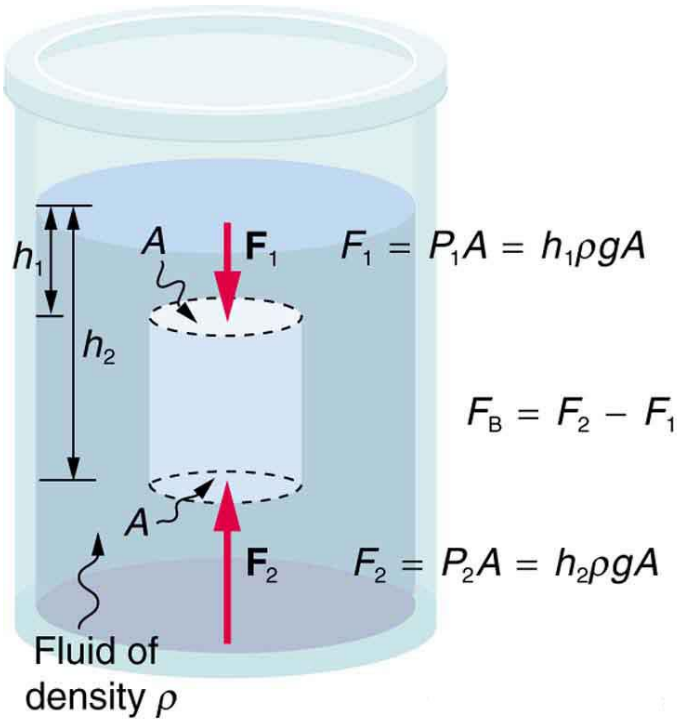
□ □ Limitations / Conditions

Archimedes' principle assumes:

Fluid is at rest

Density is uniform

No strong currents or turbulence



□ Practical Importance

Ship and submarine design

Hydrometers and lactometers

Scuba diving control

Oil-water separation

Engineering of floating platforms

□ □ Scientific Legacy

The work of Archimedes showed that:

Physical laws can be **derived from observation + reasoning**

Even simple phenomena (like water displacement) can reveal **fundamental laws**

REAL TIME EXAMPLES

Real-time examples are examples taken from **everyday life or practical situations** that you can observe happening around you **right now or in daily experience**, not just in theory.

☐ **Key Features**

- Based on **real-life situations**
- Easy to **observe and understand**
- Help connect **theory with practice**

Instead of just learning Archimedes' Principle, real-time examples help you *see* it in action:

- Blade floating on the surface of water
- Submarines float
- Mosquitoes on surface of water
- Balloons rising in air
- Stones sinking in water

These are all **real-time examples of buoyant force we will going to discuss ahead**.

☐ **Simple Line**

- "Real-time examples are practical situations from daily life that help us understand and apply scientific concepts."
- If you want, I can give real-time examples specifically for exams or make very short definitions.

Blade Floating on the Surface of Water

- ☐ **Blade Floating on the Surface of Water**

At first glance, a metal blade **should sink** (since its density is greater than water), but sometimes it **floats on the surface**. This is a classic real-life observation—and it's **not mainly due to buoyancy alone**.

- ☐ **What Actually Makes It Float?**

1. Surface Tension (Main Reason)

- Water molecules at the surface stick together, forming a **thin elastic layer**.
- This surface layer can **support light objects** like a blade if placed gently.
- The blade rests on this "film" without breaking it.

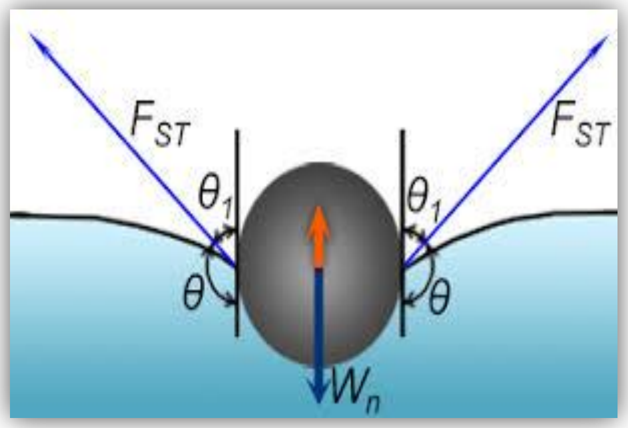
2. Role of Buoyancy (Secondary)

- A small buoyant force still acts (as per Archimedes' Principle),
- But **by itself, it is not enough** to support the blade.
- The main support comes from surface tension.

☐ Important Conditions

The blade will float only if:
It is placed **carefully and horizontally**
The water surface is **undisturbed**
No soap or detergent is present (these reduce surface tension)

☐ ☐ What Makes It Sink?



Adding soap → breaks surface tension
Dropping the blade roughly → breaks the surface film
Tilting the blade → increases pressure on one side

☐ Final Concept

☐ The blade floats mainly due to **surface tension**, not just buoyancy.
☐ This example shows that **multiple physical forces can act together** in real-life situations.

SUBMARINES FLOAT

A submarine is one of the best real-life demonstrations of **buoyant force in action** because it can **float, sink, and stay suspended**—all by controlling buoyancy.

☐ ☐ Principle Used

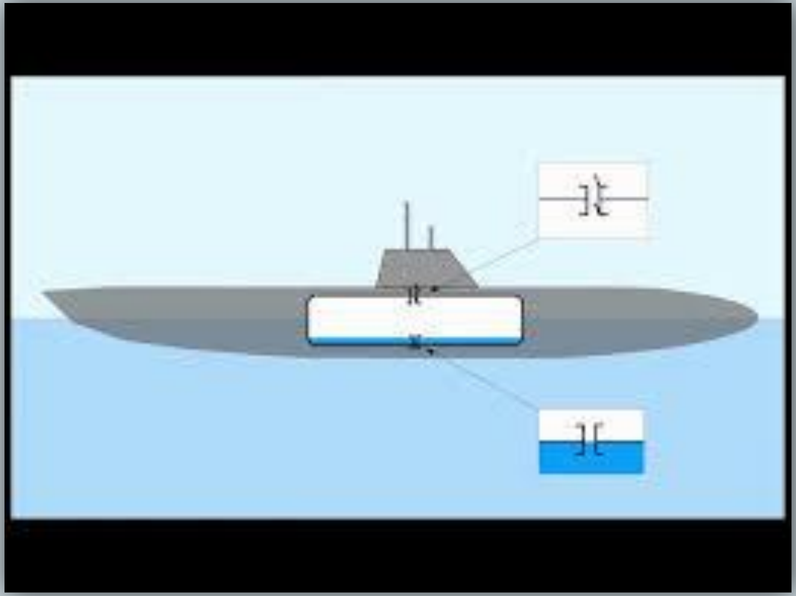
- Submarines operate according to Archimedes' Principle:
- The buoyant force on the submarine equals the **weight of the water it displaces**.

☐ How Buoyancy Works in a Submarine

- A submarine has special tanks called **ballast tanks**. By changing what's inside these tanks (air or water), it changes its **overall density**.

1. Floating on the Surface

- Ballast tanks are filled with **air**
- Submarine becomes **less dense than water**
- It displaces enough water to balance its weight
  ☐ Result: **It floats**



2. Diving (Sinking)

- Ballast tanks are filled with **water**
- Overall density increases
- Weight becomes greater than buoyant force
  ☐ Result: **It sinks**

3. Staying at a Fixed Depth (Neutral Buoyancy)

- Tanks contain a **controlled mix of air and water**
- Weight = buoyant force
  ☐ Result: Submarine remains **suspended underwater**

☐ ☐ Why This is a Perfect Example of Buoyancy

Shows all three conditions:

- Floating
- Sinking
- Neutral buoyancy

Demonstrates that **shape and density matter more than material**

Applies directly to real engineering systems

☐ Real-Life Importance

Used in naval defense and underwater exploration

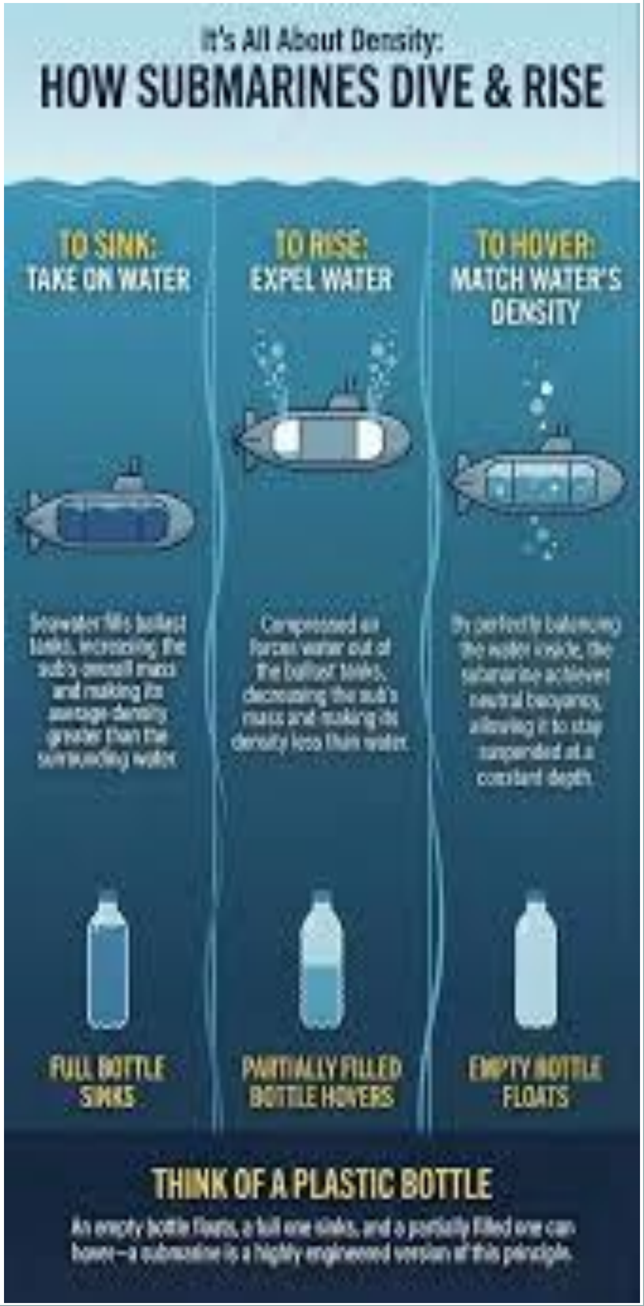
Helps scientists study deep oceans

Essential for rescue and research missions

☐ Final Concept

☐ A submarine proves that buoyancy is **controllable**

☐ By adjusting its density, it can move **up, down, or stay still** in water



Mosquitoes on Surface of Water

Mosquitoes on the Surface of Water (Real-Life Example of Buoyancy)

- Mosquitoes can rest or move on the surface of still water without sinking. This is a **natural example of buoyancy combined with surface effects of water**.

Main Principle Involved

- Their behavior is partly explained by Archimedes' Principle and strongly influenced by **surface tension.**

How Mosquitoes Stay on Water

1. ☐ Surface Tension (Main Support)

- Water molecules form a **thin elastic layer at the surface**
- This layer can support very light objects
- Mosquito legs are so light that they do not break this surface film

☐ Result: The mosquito appears to “walk” on water.

2. ☐ ☐ Role of Buoyant Force

- A small upward **buoyant force** also acts on the mosquito
- However, since mosquitoes are extremely light, this force is enough to assist in keeping them afloat
- But **surface tension is the dominant factor**

□ Special Adaptations of Mosquitoes

Long, thin legs distribute weight over a larger area
Hydrophobic (water-repelling) body helps prevent sinking
Light body mass reduces pressure on water surface

□ Real-Life Observation

Mosquitoes can:

- Rest on calm water
- Lay eggs on water surfaces
- Move without breaking the surface

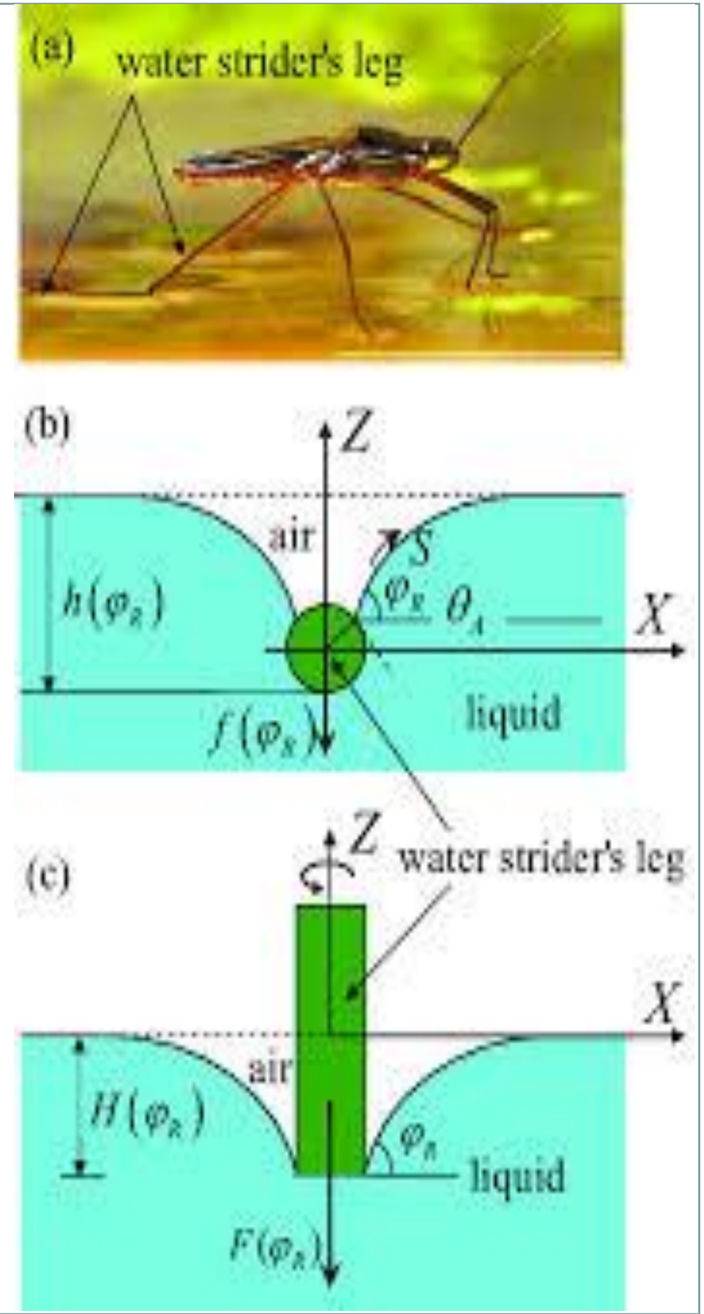
□ □ What Breaks This Effect?

Disturbing the water surface (waves, vibration)
Adding soap or detergent (reduces surface tension)
Pressing forcefully on the insect

□ Final Concept

□ Mosquitoes on water show how **surface tension and buoyancy work together in nature**

□ But the main reason they float is **surface tension, not just buoyant force**



BALLOONS RISING IN AIR

Balloons Rising in Air (Real-Life Example of Buoyancy)

- A balloon rises in air because of **buoyant force acting in gases**, not just liquids. Air behaves like a fluid, so the same principles apply.

Principle Behind It

- This is explained by Archimedes' Principle:
- An object in a fluid experiences an upward force equal to the **weight of the fluid it displaces**.
- Here, the “fluid” is **air**.

How a Balloon Rises

1. Density Difference (Main Reason)

- A balloon filled with **helium or hot air** is **less dense than surrounding air**
- It displaces heavier air around it
- Upward buoyant force becomes greater than its weight

☐ Result: It rises

2. Buoyant Force in Air

- Air pushes upward on the balloon
- If:
    - Buoyant force > weight → balloon rises
    - Buoyant force = weight → it stays suspended

☐ **Types of Rising Balloons**

☐ **Helium Balloon**

Filled with helium gas
Helium is much lighter than air
Floats upward easily

☐ **Hot Air Balloon**

Air inside is heated
Hot air expands → becomes less dense
Balloon rises due to reduced density

☐ **Real-Life Examples**

Party balloons floating in the air
Weather balloons rising to study atmosphere
Hot air balloons used for travel and tourism

☐ ☐ **Why Balloons Eventually Stop Rising**

Air becomes thinner at higher altitudes
Buoyant force decreases
Balloon reaches a height where forces balance

☐ **Final Concept**

☐ Balloons rise because **they are less dense than air**
☐ Air exerts an upward **buoyant force** on them
☐ This is a direct application of buoyancy in gases

STONES SINKING IN WATER

- A stone sinks in water because the **upward buoyant force is not enough to support its weight**.

Principle Behind It

- This is explained by Archimedes' Principle:
- A body in a fluid experiences an upward force equal to the **weight of the fluid it displaces**.

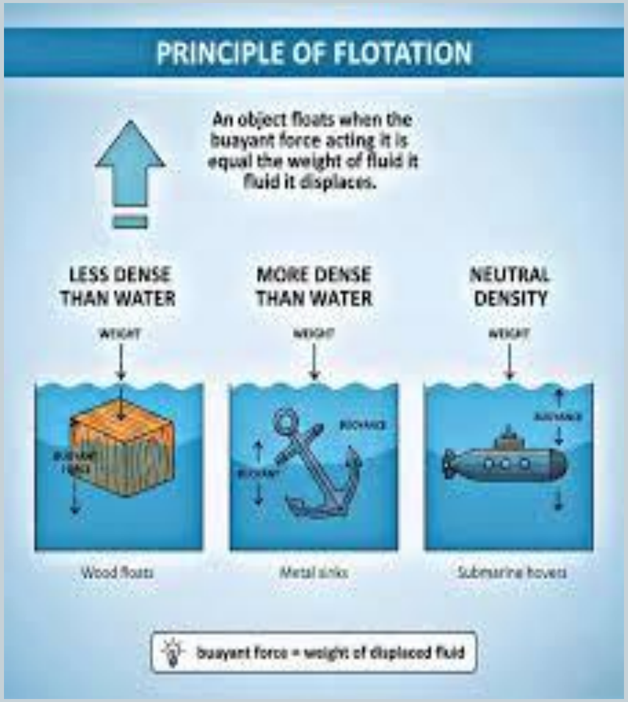
Why Stones Sink

1. High Density

- Stone has a **higher density than water**
- It is heavier compared to the amount of water it can displace
- ☐ Result: It cannot float

2. Force Balance

- When a stone is placed in water:
- Downward force (weight) is **very large**
- Upward buoyant force is **smaller**
- Weight of stone>Buoyant force
- ☐ Result: Stone sinks



☐ **What Happens in Water**

Stone goes down until it reaches the bottom

Even after full immersion, it still displaces only a limited volume of water

That displaced water is not heavy enough to balance its weight

☐ **Key Concept**

Floating or sinking depends on **density comparison**, not just size

Small but dense objects sink faster than large but light objects

☐ **Real-Life Observation**

Pebbles sink in rivers

Metal objects sink in lakes

Even small stones drop quickly in water

☐☐ **Important Insight**

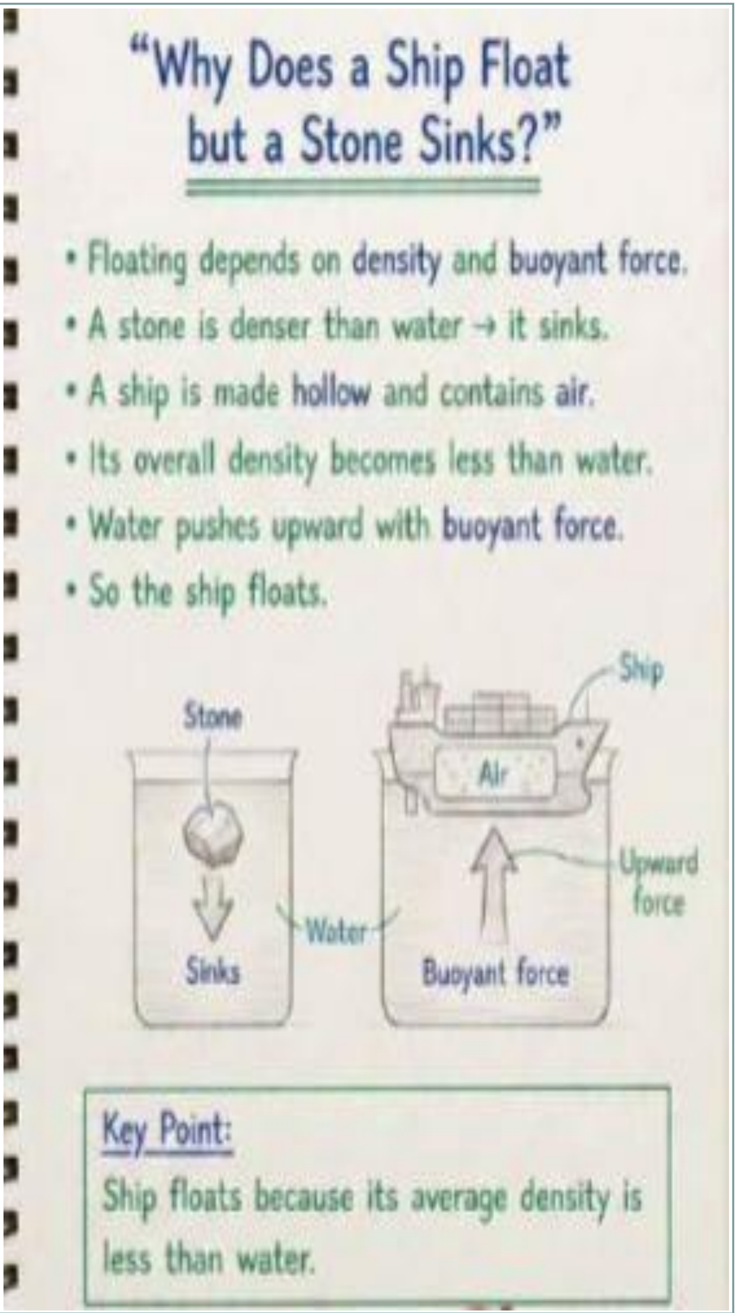
Increasing size alone does not make a stone float

Only when average density becomes less than water can an object float (like a ship)

☐ **Final Concept**

☐ Stones sink because their **weight is greater than the buoyant force**

☐ They are denser than water, so they cannot displace enough fluid to float



CONCLUSION

Buoyancy is a fundamental force in fluids that explains why objects **float, sink, or remain suspended** when placed in liquids or gases. It is based on the idea that a fluid exerts an **upward force on any object immersed in it**, equal to the weight of the fluid displaced, as described by Archimedes' Principle.

In simple terms, an object will float if the **buoyant force is greater than or equal to its weight**, and it will sink if its weight is greater. This principle depends mainly on **density and displaced volume**, not just size or material.

Buoyancy is not just a theoretical concept—it is widely seen in real life through ships floating, submarines diving, balloons rising, and objects behaving differently in water and air.

☐ **Final idea:**

Buoyancy helps us understand and predict how objects behave in fluids, making it one of the most important concepts in physics and engineering.

THANK YOU